

1(a)	A – ciliated, epithelium / epithelial tissue ; B – smooth muscle ; D – cartilage ;	3
1(b)	1 produce / release / secrete, mucus / mucin ; R excrete mucus any two from: 2 mucus forms a layer over the (ciliated) epithelium ; 3 traps / sticks to, bacteria / viruses / pathogens / microorganisms / dust / pollen / (fungal) spores / AW ; 4 mucus is, moved / AW, away from alveoli <i>or</i> lung tissue / towards back of mouth / AW, by cilia ; R 'by ciliated cells' 5 <i>idea that</i> mucus prevents (named) pathogens reaching, epithelial cells / alveoli / gas exchange surface ; A acts as a barrier to (named) pathogens	3
1(c)(i)	$pH\ 7.7 = 72(\%)$ $pH\ 7.4 = 60(\%)$;	1
1(c)(ii)	<i>allow Hb or hb for haemoglobin</i> decrease in supply of oxygen to tissues / AW ; (at lower pCO_2 / increase pH blood) affinity of, haemoglobin, for oxygen is higher or haemoglobin is more saturated <i>or a</i> A less oxygen, released / dissociates / AW, from haemoglobin (than normal) <i>must be in context of higher affinity in the tissues</i> or data quote for difference in saturation of haemoglobin in correct range of pO_2 (within 2–6 kPaO_2) ;	2